

Cody Appa, PhD

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📍 (Open to Relocation)



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cody-appa.github.io/Cody_Website/

Skills

DNA Synthesis and Assembly

- PCR, qPCR, RT-PCR, ddPCR
- Plasmid design and plasmid biology, Ligation-based DNA assembly, and molecular cloning (Benchling, NCBI)
- CRISPR-dcas12 interference vector design
- PCR primer design/troubleshooting
- DNA synthesis/assembly and scale-up

Cell Culture

- Mammalian cell culture: Cos-7 cells and Hela cells
- *E.coli* bacterial culture (
- *E.coli* transformation and genetic alteration
- Yeast cell culture
- *Chlamydia* infectious co-culture
- Maintained multiple cell lines.
- Lipid nanoparticle transfection of Cos-7 cells
- Cell banking, cryopreservation, and thawing
- Fluorescent microscopy (Nikon and Andor systems)
- Immunofluorescence, immunohistochemistry, Fluorescent in-situ hybridization

Biochemical Assays

- DNA/RNA Isolations and purification
- Protein purification and chromatography
- Bradford, ELISA, RNA-seq
- Gel electrophoresis, SDS-PAGE, Immunoprecipitation

Data Analysis/Visualization

- Proficient in Python (matplotlib/pandas) and R (ggplot2) programming languages for data analysis and visualization
- Familiar with Linux, Mac OS, and BASH

Professional Summary:

Accomplished Molecular Biologist with 8+ years of research experience in advanced gene editing technologies including CRISPR interference. Recent PhD graduate with expertise in DNA synthesis and assembly as well as plasmid biology. Published researcher with multiple peer-reviewed publications including an Editor's Pick first-author paper in mSphere. I'm excited to transition from academic research to biotech industry!

Publications:

Appa CR, Grieshaber NA, Yang H, Omsland A, McCormick S, Chiarelli TJ, Grieshaber SS. The chlamydial transcriptional regulator *Euo* is a key switch in cell form developmental progression but is not involved in the committed step to the formation of the infectious form.

August 2024

doi.org/10.1128/msphere.00437-24

Nicole A. Grieshaber, **Cody Appa**, Megan Ward, Alorah Grossman, Sean McCormick, Brendan S. Grieshaber, Travis Chiarelli, Hong Yang, Anders Omsland, Scott S. Grieshaber. The T3SS structural and effector genes of *Chlamydia trachomatis* are expressed in distinct phenotypic cell forms.

April 2024

doi.org/10.1101/2024.04.25.591156

Chiarelli TJ, Grieshaber NA, **Appa C**, Grieshaber SS. 2023. Computational modeling of the chlamydial developmental cycle reveals a potential role for asymmetric division.

March 2023

doi.org/10.1128/msystems.00053-23

Nicole A. Grieshaber, Travis J. Chiarelli, **Cody R. Appa**, Grace Neiswanger, Kristina Peretti, Scott Grieshaber. Translational gene expression control in *Chlamydia trachomatis*.

January 2022

doi.org/10.1371/journal.pone.0257259

Nicole A. Grieshaber, Justin Runac, Sierra Turner, Marissa Dean, **Cody Appa**, Anders Omsland, Scott Grieshaber. The sRNA regulated protein DdbA is involved in development and maintenance of the *Chlamydia trachomatis* EB cell form

July 2021

doi.org/10.3389/fcimb.2021.692224

Education

PhD in Molecular Biology,
Microbiology and Biochemistry
University of Idaho
August 2025 | 3.96 GPA

BS in Biology
University of Idaho
May 2019

Honors

University of Idaho College of
Science Student Research
Expo

- Best Graduate Student
Poster

-March 2025

Max and Sharon Walker
College of Science Scholarship
-November 2024

Editor's Pick First Author
Publication

- [https://doi.org/10.1128/
msphere.00437-24](https://doi.org/10.1128/msphere.00437-24)

-August 2024

Chlamydia Basic Research
Society International
Conference

- Jane Raulston Award for
Best Graduate Student
Poster

-March 2023

Presentations

Idaho Student Research
Exposition

-March 2025

INBRE Statewide Research
Conference

-July 2024

Chlamydia Basic Research
Society Conference

-March 2023

American Society of
Microbiology Northwestern
Branch Conference.

-November 2022

References

Available on request.

Work Experience:

Research Assistant/PhD Candidate:

August 2021 - August 2025

- Method Development and Innovation: 8+ years of research experience; developing and driving molecular biology assays and DNA manipulation techniques. Culminated in discovering the Saturn Body, a transitional cell form in *Chlamydia trachomatis* development.
- Advanced DNA Manipulation & Synthesis: Designed and executed complex DNA synthesis and cloning projects including designing, engineering, and cloning 100+ unique plasmids. Transformed plasmids into *E.coli* for DNA scale-up and *Chlamydia* transformation for genetic assays including CRISPR-interference plasmids.
- Cell Culture and Aseptic Technique: 8+ years of mammalian and bacterial cell culture experience. Worked in a Biosafety 2+ environment. Managed multiple concurrent cell lines with simultaneous infectious co-culture with zero cross contamination.
- Independent Project Leadership: Led multi-year research initiatives from experimental design through data analysis and publication, consistently meeting project milestones while managing multiple concurrent studies.
- Technical Documentation & Communication: Maintained good documentation practices and kept detailed experimental records, authored research publications, and presented findings to diverse scientific audiences.
- Quantitative Analysis & Bioinformatics Tools: Applied programming languages Python (matplotlib, NumPy, Pandas) and R (ggplot2) for experimental data analysis, sequence alignment, and statistical modeling, bridging wet lab work with data science.
- Mentorship & Teaching: Trained multiple undergraduate researchers in advanced molecular biology techniques, demonstrating ability to share technical expertise and develop others.
- Cross-Functional Collaboration and Communication: Collaborated with cross-functional teams and departments. Multiple award-winning presentations and an Editor's Pick first author publication demonstrate ability to effectively communicate complex data.

Research Technician:

August 2019 – May 2021

- Research Experience: Created promoter reporter vector constructs for multiple genes of interest. Cell-based and plate-based assays and plate readers.
- Lab Maintenance and Upkeep: Equipment maintenance of lab tools and assisted with orders and reagent prep.

Undergraduate Research Assistant:

January 2017 – May 2019

- Research Experience: Developed E- riboswitch repression vectors. Transformed plasmids into *E. coli* and *Chlamydia*.